Introduction to Statistical Modelling

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1 Statistical Modelling

Variability is all around

- Data obtained when performing any experiment show variability.
- Statistics is a discipline that has been developed in response to the experimenters whose data exhibit variability.

Random sampling

Population → sample

$X \sim \text{MODEL}(\theta)$

θ Parameter(s) unknown

$\boldsymbol{X} = (X_1, \ldots, X_n)$

Descriptive statistics

$t(\boldsymbol{X})$ function of the data

Inference: Estimation and Hypothesis testing

We use $t(\boldsymbol{X})$ to explain θ

Statistical Modelling.

- A mere description of the data is useful, but proper science needs to know why we have observed those data and to do so, we use a MODEL.
- In general, a model is a small-scale representation of reality:
  - either a mere physical representation,
  - a description of reality,
  - a tool to understand the reality or
  - a tool for predicting future behavior.
- The best feature of a model is to be as accurate as possible in their task of representing reality.
- Statistics allows us to incorporate the variability present in real life in our models through randomness.
- But more importantly, it allows us to predict.
- Still: "essentially, all models are wrong, but some of them are useful" (Box).
- "The problem formulation is more essential than its own solution, which may simply be a mathematical or experimental skill" (Einstein).

Formulation of a model.

In order to properly desing a model for a particular real problem:

- To understand the physical background of the problem. Statisticians often work in conjunction with specialists from different areas and need to know the basics of the problem area.
- Clearly understand the objective. It is not an easy task to achieve it.
- Be clear about what the person who has consulted us wants. No sense in making complex analysis if their interest is lower.
- Put the problem in statistical terms! This is the key step in which sometimes some irreparable mistakes are made. But once it is in statistical terms, the solution is usually routine.
- But beware, having a model to read and process the data is not enough if we are not able to draw conclusions.

Basics of modelling.

Some well known definitions:

- Response Variable $\longrightarrow$ Characteristic of interest.
- Population: set of values of the response variable that we would get if the process of data collection is repeated indefinitely.
- The variable of interest has a probability distribution associated, population distribution (and is what usually stands for *population*).
- Parameters are the characteristics of interest of the population (mean, variance, population proportion, etc.).
- A random sample of observations of the variable X of sample size n is a set of random variables $X_1, X_2, \ldots, X_n$ independent and identically distributed with the same distribution of the variable X.
- Statistic: data reduction that allows us to perform inference (is a function of the sample observations). For instance, the sample mean, the variance.

Modelling: types of variables.

- When modelling a real problem:
  - what do we want to explain?
  - and based on what?
- This classifies variables in:
  - Response variable: the one we want to explain
  - independent (or predictives or explicatives): those we use to explain the response
- Variables can also be classified by their attribute:
  - Qualitative $\longrightarrow$ Non numeric (categorical)
    - ⋆ Nominal (no order): sex, presence/absence, type of fish (cod, hake, etc.)
    - ⋆ Ordinal (ordered values): rating of the fishing area (poor, fair, good)
  - Quantitative $\longrightarrow$ numeric
    - ⋆ Discrete (finite or numerable quantity): number of species
    - ⋆ Continuos (values in real numbers): weight of the fish

Statistical models.

- Most statistical models have a structure like:
  - Response Variable to be explained.
  - A systematic component that contains the "general" information of the system under study and it is expressed as a combination of explanatory variables in the form of a parametric equation.
  - A random component reflecting the inherent variability in each particular situation.
- The response variable can be distributed by means of a Gaussian distribution (Linear models) ...
- ... or by any other distribution belonging to the Exponential family (Generalized linear models: Logistic regression, Poisson regression, Beta regression, etc.).

Describing the systematic component

- Depending on the type of variable, the explanatory are classified as:
  - Qualitative $\longrightarrow$ Factors (with their corresponding "levels")
    - Fixed effects (if levels are previously fixed: Sex)
    - Random effects (if the levels are a random sample of the possible levels: ship)
  - Quantitative $\longrightarrow$ Covariates: bathymetry, temperature, etc.
- Often the systematic component is expressed as a linear combination (but can also be non-linear).

Linear models

- If the response variable is normal (most cases) and the relationship is linear, we have a Linear model.
- Example: when we want to describe the age of a fish (RV) depending on its lenght and weight (covariates), that is,

$$\text{Age}_i \sim N(\mu_i, \sigma^2)\,;$$

$$\mu_i = \beta_0 + \beta_1 \text{lenght}_i + \beta_2 \text{weight}_i\,.$$

- When the explanatory variables are fixed effect factors, "dummy" variables are introduced.
- For instance, including sex as a factor:

$$\text{Age}_i \sim N(\mu_i, \sigma^2)\,;$$

$$\mu_i = \beta_0 + \beta_1 \text{lenght}_i + \beta_2 \text{weight}_i + \beta_3 \text{female}_i\,.$$

Most common used statistical models.

- Generalized linear models, when we want to describe the ocurrence (presence/absence) of a species or the number of species or the proportion of discards depending on some covariates.
- Generalized linear mixed models, when we also include a random effect in the model.
- Generalized additive models, when the relationship with the covariate is not linear.
- Time series models, when there is temporal autocorrelation.
- Spatial models, when there is spatial autocorrelation.
- Spatio-temporal models, when there is spatio-temporal autocorrelation.

Most common used statistical models (2).

Explanatory Variable(s)	Response variable errors (unique)			
	NORMAL	BINOMIAL	POISSON	OTHER
There is no	t-test 1s	Est. H.T. π	Est. H.T. λ	Est. H.T.
Binary or Discrete	t-test 2s ANOVA 1f	Log. Reg. Log-Lin	PoissonReg. GLM	GLM GLM
Discretes	ANOVA mf	GLM	GLM	GLM
Continuos (s)	Linear Reg. Multiple Reg.	S. Log. R. M.Log.R.	S. Poisson Reg. Poisson Reg.	GLM GLM
Continuos and discretes	Covariance Analysis	Generalized Log. Reg.	Generalized Poisson Reg.	GLM GLM
+ R. effects	LMM	GLMM	GLMM	GLMM
+ f(Cont.)s	AM	GAM	GAM	GAM
+ effects + f()s	AMM	GAMM	GAMM	GAMM
+ Sp. effects	S LMM	S GLMM	S GLMM	S GLMM
+ temp. effects	S-T LMM	S-T GLMM	S-T GLMM	S-T GLMM

2 Bayesian approach to Linear Models

Bayesian Linear models

- A linear model with covariates, fixed-effect factors (using dummy variables) and interactions has the form:

$$
\begin{aligned}
Y_i &\sim N(\mu_i, \sigma^2)\,; \\
\mu_i &= \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} + \\
&\quad + \gamma_1 d_{i1} + \ldots + \gamma_{q-1} d_{iq-1} + \\
&\quad + \delta_{11} x_{i1} d_{i1} + \ldots + \delta_{1,q-1} x_{i1} d_{iq-1} + \\
&\quad + \ldots + \\
&\quad + \delta_{p1} x_{ip} d_{i1} + \ldots + \delta_{p,q-1} x_{ip} d_{iq-1} \quad \forall i = 1, \ldots, n.
\end{aligned}
$$

- Objective: estimate the parameters $(\beta_0, \ldots, \beta_p, \gamma_1, \ldots, \gamma_{q-1}, \delta_{11}, \ldots, \delta_{p,q-1}, \sigma^2)$.

Bayesian Linear models (2)

- In order to make inference in this model we can use Least Squares estimation (Classical) or use the Bayesian approach. In this latter case:
  - Specify the relationship between the mean and the covariates,
  - determine the likelihood function of the data,
  - specify the priors for the parameters,
  - combine prior information with the likelihood and obtain the posterior distribution of the parameters.
- The posterior distribution is a probability distribution (we can use it in terms of probability) that contains all the information about the parameters.
- The problem for many years, not now, is that these posteriors usually do not have analytical expressions. In the case of linear models are nearly analytical ...

Bayesian Linear models (3)

- The information gathered from the experiment

$$y_i \sim N(\mu_i, \sigma^2) \qquad i = 1, \ldots, n$$

and the relationship between the response and the covariates $\mu_i = \boldsymbol{X\beta}$ can be expressed through the likelihood:

$$l(\boldsymbol{\beta}, \sigma) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}(y - \boldsymbol{X\beta})'(y - \boldsymbol{X\beta})\right\}$$

- Using a non-informative (improper) prior for the parameters

$$p(\boldsymbol{\beta}, \sigma) \propto \frac{1}{\sigma^2}$$

the resulting (analytical) posterior is

$$p(\boldsymbol{\beta}, \sigma^2 | \boldsymbol{y}, \boldsymbol{X}) = p(\boldsymbol{\beta} | \boldsymbol{y}, \boldsymbol{X}, \sigma^2) p(\sigma^2 | \boldsymbol{y}, \boldsymbol{X})$$

where

$$\begin{aligned} p(\boldsymbol{\beta} | \boldsymbol{y}, \boldsymbol{X}, \sigma^2) &= N_k(\hat{\boldsymbol{\beta}}, (\boldsymbol{X}'\boldsymbol{X})^{-1}\sigma^2) \\ p(\sigma^2 | \boldsymbol{y}, \boldsymbol{X}) &= \text{Inv-}\chi^2(n-k, \hat{\sigma}^2) \end{aligned}$$

and being $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}\boldsymbol{y}$ and $\hat{\sigma}^2 = \frac{1}{n-k}(y - \boldsymbol{X\beta})'(y - \boldsymbol{X\beta})$.

3 Bayesian approach to GLM's

Reminder of GLM's

- Generalized Linear Models (GLM) extend the linear regression model in order to accommodate:
  - non-normal responses, e.g. binomial data, frequency data, etc.
  - and transformation to linearity
- Well known models are logistic regression, log-linear models for frequency tables, Poisson regresion, Gamma regresion, Beta regression, etc.
- The conceptual advantage is that many real situations with non-normal data are reduced to regression modelling.
- Example. A model that explains the captures (positive, not Gaussian distributed) of a species by means of chlorophyll and bathymetry:

$$\text{CPUE}_i \sim \mathit{Gamma}(\mu_i, \phi)\,;$$

$$\mathit{log}(\mu_i) = \beta_0 + \beta_1 \text{CL}_i + \beta_2 \text{BAT}_i\,.$$

Reminder of GLM's (2)

A Generalized linear model consists of:

- A set of $Y_1, \ldots, Y_n$ random variables (response) independent and identically distributed inside the Exponential family.
- A set of explanatory variables $X^{(1)}, X^{(2)}, \ldots, X^{(p)}$ that along with a parametric vector $(\beta_0, \beta_1, \ldots, \beta_p)$ form the linear predictor:

$$\eta_i = \boldsymbol{X_i}^t \boldsymbol{\beta} = \beta_0 + \beta_1 X_i^{(1)} + \ldots + \beta_p X_i^{(p)},\ i = 1, \ldots, n$$

- A monotonic and differentiable function called *link function* $g()$, defining the relationship between the mean of the response $\mu_i = E(Y_i)$ and the linear predictor:

$$g(\mu_i) = \eta_i$$

- Equivalently: $\mathrm{E}[Y_i] = g^{-1}(\beta_0 + \beta_1 X_i^{(1)} + \ldots + \beta_p X_i^{(p)})$.

Inference in GLM's.

- Starting with the corresponding likelihood which contains the available information about the parameters: the coefficients of the linear predictor.
- For example, in the case of Poisson data with mean μ_i and logarithm link $g(\mu_i) = \log(\mu_i)$, the likelihood is:

$$\ell(\boldsymbol{\beta}|\mathrm{y}, \boldsymbol{X}) = \prod_{i=1}^{n} \left(\frac{1}{y_i!}\right) \exp\{y_i \boldsymbol{X_i}^t \boldsymbol{\beta} - \exp(\boldsymbol{X_i}^t \boldsymbol{\beta})\}$$

- In classical statistics: need of numerical methods as the likelihood does not have an "easy" expression. Maximum likelihood estimates are found using an iteratively reweighted least squares algorithm based on a Newton–Raphson method, namely the Fisher's scoring method.
- The result consists on a vector of the estimated parameters: $(\hat\beta_0, \hat\beta_1, \ldots, \hat\beta_p)$ along with the sampling distribution of each parameter which allows hypothesis testing and confidence intervals.
- Find the best model by comparing AIC or Deviance criterions and validate the final model by analising residuals.
- But, what about the Bayesian approach? Not much different conceptually.

Bayesian analysis in GLM's.

- Conceptually the Bayesian specification is "straightforward".
- Starting with the corresponding likelihood, ...
- ... we "only" need to assign a prior for regression coefficients.
- But, how to do this assignment? Not an easy answer, could talk long about this!
- Easy option: to choose conjugate or non-informative independent priors. For instance normal or flat:

$$p(\beta|\text{data}) \propto 1 \times \prod_{i=1}^{n} \left(\frac{1}{y_i!}\right) \exp\{y_i \boldsymbol{X_i}^t \beta - \exp(\boldsymbol{X_i}^t \beta)\}\,.$$

- But, as usual in Bayesian modelling, there are no closed form solution available for the posterior distribution of parameters.
- As usual, once the inference is done we need to find the best model by selecting among the possible covariates to be included in the model using *DIC* criterion.

4 INLA and latent Gaussian models

Laplace approximations

- Laplace approximations for the numerical integration are a good way to solve the integration problem in Bayesian Statistics.
- Nevertheless, they have the dimensionality problem. They behave fine for a small number of dimensions, but actually all the practical situations involve many dimensions (a large large number of parameters).
- The INLA[1] approach is a Laplace approximation method integrated in a very particular type of models (that includes the vast majority of models we work with regularly). And they do not have the problem of dimensionality.

[1]https://r-inla.org/

Integrated Nested Laplace Approximation: INLA.

- Although in theory MCMC is always possible to implement, this approximation has some backwards:
  - in terms of both convergence and computational time and
  - the implementation itself (that can be very tricky for those non experts in programming).
- INLA appears as an option to perform a numerical approximation much faster for a particular kind of models, latent Gaussian models (LGM).
- Latent Gaussian models are an apparently simple but very flexible construct in statistical applications which covers a wide range of common statistical models spanning from (generalized) linear models, (generalized) additive models, smoothing spline models, state space models, semi-parametric regression, spatial and spatiotemporal models, log-Gaussian Cox processes and geostatistical and geoadditive models.
- By its own construction (hierarchical structure in three stages), latent Gaussian models are a class of hierarchical Bayesian models.
- We only need then (although it is not that easy always) to rewrite a model in that hierarchical form in order to be considered as a latent model.

Hierarchical structure of a latent Gaussian model.

1. The first stage is formed by the conditionally independent likelihood function:

$$p(\boldsymbol{y}|\boldsymbol{\theta}, \boldsymbol{\psi}) = \prod_{i=1}^{n_d} p(y_i|\eta_i(\boldsymbol{\theta}), \boldsymbol{\psi}),$$

   where:
   - $\boldsymbol{y} = (y_1, \ldots, y_{n_d})^T$ is the response vector,
   - $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)^T$ is the latent field,
   - $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_m)^T$ is the hyperparameter vector and
   - $\eta_i(\boldsymbol{\theta})$ is the i-th linear predictor that connects the data to the latent field.

2. The second stage is formed by the latent Gaussian field, where we attribute a Gaussian distribution with mean $\mu(\boldsymbol{\psi})$ and precision matrix $Q(\boldsymbol{\psi})$ to the latent field $\boldsymbol{\theta}$ conditioned on the hyperparameters $\boldsymbol{\psi}$, that is:

$$\boldsymbol{\theta}|\boldsymbol{\psi} \sim N(\mu(\boldsymbol{\psi}), Q^{-1}(\boldsymbol{\psi}))$$

   In most cases, the latent Gaussian fields have sparse precision matrix $\boldsymbol{Q}(\boldsymbol{\psi})$, i.e., they are Gaussian Markov Random Fields (GMRF).

3. Finally, the third stage is formed by the prior distribution assigned to the hyperparameters:

$$\boldsymbol{\psi} \sim p(\boldsymbol{\psi})$$

5 Statistical models with random effects

GLMMs: models with independent random effects

- A factor with random effects is a factor which levels are a random sample of the possible levels.
- They allow us to easily incorporate extra variability caused by each of those different levels of the factor.
- Example. A model that also incorporates the effect of the ship when we want to explain the captures by means of chrolophylla and bathymetry as already discussed:

$$\text{CPUE}_i \sim Ga(\mu_i, \phi)\,;$$

$$log(\mu_i) = \beta_0 + \beta_1 \text{CL}_i + \beta_2 \text{BAT}_i + v_{j(i)}\,;$$

- $v_{j(i)}$ represents the extra variability added in the model by vessel j, expressed as

$$v_{j(i)} \sim \text{N}(0, \sigma_v^2)\,.$$

GLMMs as LGMs.

The previous example can also be seen as a Latent Gaussian Model and so analyzed with `R-INLA`

1. captures follow a conditionally independent Gamma likelihood function:

$$CPUE_i \sim \text{Gamma}(\mu_i, \phi),\ i = 1, ..., n_i\,.$$

2. we also assume that

$$\begin{array}{rcl} \eta_i & = & log(\mu_i) = \beta_0 + \beta_1 \text{CL}_i + \beta_2 \text{BAT}_i + v_{j(i)} \\ v_{j(i)} & \sim & N(0, \sigma_v^2), \\ \boldsymbol{\beta} & = & [\beta_0, \beta_1, \beta_2]^T,\ \beta_i \sim N(0, \sigma_\beta^2),\ \sigma_\beta^2 \text{ known} \end{array}$$

   So, in this case, $\boldsymbol{\theta} = (\beta, v_1, ..., v_{n_j})$ is the latent Gaussian field (note that a Gaussian prior is assigned for each element of the latent field, so that $\boldsymbol{\theta}|\boldsymbol{\psi}$ is Gaussian distributed).

3. We can assing the prior of $\boldsymbol{\psi} = (\phi, \sigma_v^2)$.

GAMMs: models with splines (the INLA way)

- GLMMs with independent random effect do not cover non-linear relationships between the response variable and the covariates.
- In INLA, we can do this by considering those relationships as random walks of order 1 and 2, and so, being a latent Gaussian model.
  - First order Random Walk (RW1)

$$\Delta x_j = x_j - x_{j+1} \sim \mathrm{N}\left(0, \sigma^2 = \frac{1}{\tau}\right)$$

  - Second order Random Walk (RW2)

$$\Delta^2 x_i = x_i - 2x_{i+1} + x_{i+2} \sim \mathrm{N}\left(0, \sigma^2 = \frac{1}{\tau}\right)$$

  - The last stage is the prior for the hyperparameter τ, usually reparametrized in terms of its logarithm.

Example: Smoothing time series of binomial data.

The number of occurrences of rainfall over 1 mm in the Tokyo area for each calendar year during two years (1983-84) are registered. It is of interest to estimate the underlying probability p_t of rainfall for calendar day t which is, a priori, assumed to change gradually over time. For each day $t = 1, ..., 366$ of the year we have the number of days that rained y_t and the number of days that were observed n_t.

1. A conditionally independent binomial likelihood function:

$$y_t|\eta_t \sim \mathrm{Bin}(n_t, p_t), \ \ t = 1, ..., 366$$

with (usual) logit link function:

$$p_t = \frac{\exp(\eta_t)}{1 + \exp(\eta_t)}$$

2. We assume that (instead of a linear predictor), $\eta_t = f_t$, where f_t follows a circular random walk of second order (RW2) model with precision τ:

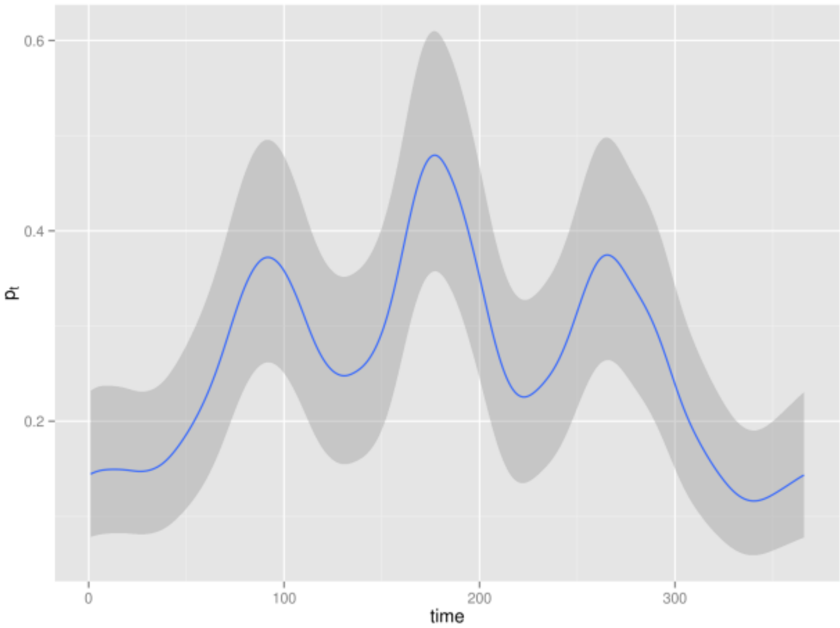
$$\Delta^2 f_i = f_i - 2f_{i+1} + f_{i+2} \sim N(0, \tau^{-1}).$$

The fact that we use a circular model here means that in this case f_1 is a neighbor of f_{366}. So, in this case $\boldsymbol{\theta} = (f_1, ..., f_{366})$ and again $\boldsymbol{\theta}|\psi$ is Gaussian distributed.

3. To assing the prior of $\psi = (\tau)$:

$$\log(\tau) \sim \mathrm{logGamma}(1, 5 \cdot 10^{-5}) \ .$$

Example: Smoothing time series of binomial data.



Posterior mean of p_t together with a 95 % credible interval.

Model selection scores in `R-INLA`

When we use different covariates and random effects, we need some measures to select the best model:

- **DIC**: within-sample predictive score

$$DIC = 2 * E(D(\boldsymbol{\theta})) - D(E(\boldsymbol{\theta}))\,.$$

- **WAIC**: within-sample predictive score

$$WAIC = \sum_i var_{post}(log(p(y_i|\boldsymbol{\theta})))\,.$$

- **LCPO**: leave-one-out cross-validation score

$$CPO_i = p(y_i|y_{-i})\,,$$

$$LCPO = -\overline{log(CPO_i)}\,.$$

6 References

References

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